FOUNDER DOSSIER · 2026

Sushanth Paatnaik

Inventor, deep-tech founder, and six-time Indian Presidential awardee. Engineering matter, capital, and scale across graphene, nano-materials, and industrial systems — from India for the world.

SIX PRESIDENTIAL AWARDS · TWENTY-SEVEN HONORS OF RECORD
TWENTY-THREE GRAPHENE INNOVATIONS · SIX OPERATING VENTURES
BHUBANESWAR · NEW DELHI · AHMEDABAD

FOREWORD

An inventor, quietly building industrial futures.

Sushanth Paatnaik is an inventor, deep-tech founder, and six-time Indian Presidential awardee. His work spans graphene and advanced materials, industrial commercialization, and the patient capital that carries frontier science from the lab into operating reality.

Born in Bhubaneswar, Odisha, the work began in a borrowed workshop at fourteen — a breath-powered wheelchair built for a man with locked-in syndrome. That afternoon set the brief: not optimisation, but agency. Performance is the means; agency is the brief.

Fifteen years on, that brief now spans six operating ventures, twenty-three graphene innovations from bench to field, and a register of recognition that includes three Presidents of India, MIT TR-35, NASA, TED-India, and BRICS.

> *Engineer matter. Engineer capital. Engineer scale.*
> *Hold all three in one hand long enough for the work to outlive the inventor.*

27 HONORS OF RECORD	06 PRESIDENTIAL AWARDS	60+ KEYNOTES SINCE 2010
23 GRAPHENE INNOVATIONS	06 OPERATING VENTURES	14+ YEARS OF RESEARCH

CHAPTER 01

Origin

A workshop, a wheelchair, and a question. The first decade — from a school-bench prototype in Bhubaneswar to IISER Bhopal, NASA, and the long apprenticeship of an inventor.

01 · ORIGIN

A workshop, a wheelchair, and a question.

PORTRAIT · BHUBANESWAR, INDIA

At fourteen, in a workshop borrowed for a weekend, Sushanth built a breath-powered wheelchair for a man with locked-in syndrome who could no longer ask for water. That afternoon rewrote what engineering meant: not optimisation, but agency.

Every venture since has begun from the same question — who is this for, and what does dignity look like for them at scale?

Academia · IISER Bhopal · OCT Bhopal

BSc. from IISER Bhopal under the KVPY-SP scholarship — chemistry, physics, and the discipline of asking questions matter cannot easily answer.

BEd. (ETE) from OCT, Bhopal under the Special Achiever category — the conviction that an inventor who cannot teach has not really finished the invention.

> *The work began in a borrowed workshop at fourteen — and has not really paused since.*

From inventor to ecosystem

The first decade was about inventions — ten-plus working prototypes shipped from a borrowed workshop. The second is about systems — companies, capital, and the talent that compounds across them. The third, by design, will be about handing the work to the next generation of Indian builders who never had to ask permission.

CHAPTER 02

Material Intelligence

One material platform. Twenty-three industrial expressions. A private R&D archive of graphene engineering — calibrated for concrete, solar, batteries, water, hydrogen, mobility, storage and armour.

02 · MATERIAL INTELLIGENCE · SUBSTRATE

The substrate beneath everything.

A single material platform — monolayer carbon, sp^2 bonded, 0.142 nm apart — engineered into twenty-three industrial expressions and traced from bench formulation through plant pilot to field deployment.

1 ATOMIC LAYER · MONOLAYER CARBON	130 GPa TENSILE · ~200× STEEL	2,630 M^2/G SURFACE · PER GRAM
5,000 W/MK · ~13× COPPER		

Commercial · Field-Deployed

CONSTRUCTION · CEMENT · PATENT

Graphacrete

49.5 MPA · −40 KG/M^3 CEMENT

Graphene nano-platelet admixture transforming standard concrete into a high-performance material.

SOLAR · COATINGS · PATENT

Graffisol

+10–12% ANNUAL YIELD · SELF-CLEANING

Solar coating delivering higher annual yield, panel cooling and superhydrophobic self-cleaning surfaces.

CERAMICS · COATINGS · RETAIL

Ceraphene

9H+ HARDNESS · ONE-THIRD PREMIUM PRICE

Graphene-enhanced ceramic coating with 9H+ hardness at a fraction of the cost of premium alternatives.

MOBILITY · COMBUSTION · FLEET TRIAL

Ignitron D

25% EFFICIENCY · 20% LOWER EMISSIONS

Graphene-enhanced diesel combustion optimization for industrial fleets, logistics systems, and heavy-duty engines.

Additional commercial products: **HD-G-PE** (graphene masterbatch, +30% tensile, 100× barrier) and **Graphenodes** (next-gen graphene polymer cathode/anode for high-density batteries).

02 · MATERIAL INTELLIGENCE · PILOT & R&D

Plant pilots and bench formulations.

WATER · RECOVERY · PILOT

Aquamax

95%+ RECOVERY · 12–24 MO ROI

World-first hybrid HAMR + HGMC system recovering 95%+ of cooling tower plume water.

THERMAL POWER · COMBUSTION · PLANT PILOT

Coalorix

15% COAL UTILIZATION · 35% LOWER EMISSIONS

Nano-engineered coal combustion optimization for thermal plants, industrial furnaces, and energy infrastructure.

TRIBOLOGY · LUBRICANTS · INDUSTRIAL PILOT

Lubritron

UP TO 6% FUEL SAVINGS · 40% WEAR REDUCTION

Nano-enabled molecular engine oil additive for all engine types — improving efficiency, reducing wear, extending oil life.

GRID STORAGE · STACK PILOT

Voltaphene

GRID-SCALE STORAGE

Graphene-enabled energy storage systems for grid and mobility applications.

Additional pilots

- Ignitron P — Petrol combustion, 15% efficiency improvement.
- Rustene — Graphene anti-corrosion shield for steel, marine and industrial assets.
- Gryogen — Hydrogen-selection membrane for clean fuel production.
- Mariphene — Low-energy graphene desalination membrane.
- Aerophenter — Atmospheric water harvesting via graphene surfaces.
- Fibrasphene — Reinforced graphene glass fibres for stronger composites.

R&D · Bench

- Armophene — Next-gen graphene ballistics: lighter, stronger armour.
- Hydrocell — Hydrogen fuel cell stack for clean power.

- Bitumax — Bitumen additive extending pavement life 1.5–2×.
- Pyronex — Multi-functional coating: fire, heat, UV, microbe shield.
- Graphyre — Graphene performance tyres for grip and mileage.
- Graphosite — Ultra-light, ultra-strong structural composites.
- Thermaphene — Smart thermal fabrics with active temperature regulation.

CHAPTER 03

Industrial Translation

Frontier science only matters when it reaches the industrial world. Five ways the desk opens for collaboration, advisory, and partnership.

03 · INDUSTRIAL TRANSLATION

Five ways the desk opens.

A single selective desk for collaboration across industry, research, ventures, speaking, and global partnerships. Invitation-based, mandate-led, designed to move frontier materials further into the industrial world.

01

INDUSTRIAL COLLABORATION

Materials, infrastructure, manufacturing systems.

Co-development with incumbents integrating graphene, advanced coatings, and nano-additives into existing product lines and industrial infrastructure — without re-engineering supply chains.

FIT · MANUFACTURERS, MATERIALS BUYERS, OPERATORS · HORIZON · 12–36 MONTHS

02

RESEARCH & DEEP-TECH

R&D direction, applied science, lab collaboration.

Embedded research direction with universities, national labs, and corporate R&D groups working on materials, energy storage, water systems, and climate infrastructure.

FIT · UNIVERSITIES, NATIONAL LABS, CORPORATE R&D · HORIZON · 24–60 MONTHS

03

VENTURE & STRATEGIC ADVISORY

Commercialization, innovation systems, venture architecture.

Founder-to-founder advisory, selective board seats, and capital co-architecture for deep-tech ventures at the materials, climate, and industrial interface.

FIT · FOUNDERS, OPERATORS, BOARDS, STRATEGIC LPS · HORIZON · MULTI-YEAR VEHICLES

04

SPEAKING & THOUGHT LEADERSHIP

Keynotes, institutional stages, editorial voice.

Keynotes, long-form interviews, panels, and stage conversations on invention, deep-tech, the carbon century, and India's industrial trajectory.

FIT · CONFERENCES, MINISTRIES, PUBLICATIONS · HORIZON · QUARTERLY BY INVITATION

05

GLOBAL PARTNERSHIPS

Academia, climate systems, cross-border collaboration.

Cross-border partnerships with academic institutions, climate organisations, and industrial transition programmes across India and the international ecosystem.

FIT · ACADEMIA, CLIMATE BODIES, SOVEREIGN, MULTILATERAL · HORIZON · MULTI-YEAR

Building systems for industrial futures and material intelligence. When the work is right, the desk opens.

CHAPTER 04

Recognition

Twenty-seven honors of record. Three Presidents of India. Six citations. From a school-bench prototype to Rashtrapati Bhavan, MIT, NASA, TED, and Silicon Valley.

04 · RECOGNITION · HALL OF FAME

A continuous reel of recognition.

Presidential demonstrations, mainstage keynotes, diplomatic honours, and institutional citations — a photographic archive of two decades on stage, in the lab, and in conversation with institutions.

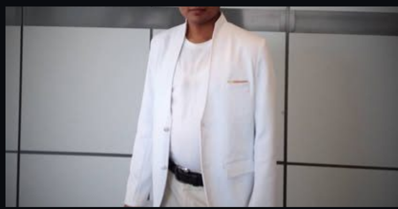
WITH DR. A.P.J. ABDUL KALAM · 2008

WITH SMT. PRATIBHA PATIL · 2009

WITH SHRI PRANAB MUKHERJEE · 2013

TED-INDIA · BANGALORE · 2012

NASA · KENNEDY SPACE CENTER · 2011



MIT TR-35 · TECH REVIEW · 2010

INTEL IRIS · BEST POPULAR INVENTION · 2010

SILICON VALLEY ADDRESS · 2024

BRICS ROUNDTABLE · NEW DELHI · 2022

STARTUP20 · G20 INDIA · 2023

EMBASSY OF ITALY · H.E. BARTOLI

SHRI NITIN GADKARI · TRANSPORT BHAWAN

Twenty-seven honors of record.

Defining honors

- Six-Time Indian Presidential Awardee (2008–2013) — Three Presidents of India: Dr. A.P.J. Abdul Kalam, Smt. Pratibha Patil, Shri Pranab Mukherjee · National Innovation Foundation.
- MIT TR-35 Awardee (2010) — Among the world's leading young innovators under 35 · MIT Technology Review.
- Intel IRIS National Recognition (2010) — Best Popular Invention · the breath-operated wheelchair.
- NASA Recognition (2011) — Kennedy Space Center / U.S. Space & Rocket Center, Huntsville.
- TED-India Speaker (2012) — Among the youngest speakers ever featured.
- MIT Fab-10 & Fab-11 (2013–14) — Center for Bits and Atoms.

Achievement ledger (selected)

- 2008 — Presidential Award (A.P.J. Abdul Kalam, NIF); Jawaharlal Nehru & CBSE National Science Exhibition Awards.
- 2009 — Presidential Awards (Pratibha Patil & A.P.J. Abdul Kalam, NIF); KVPY Fellow; NCSC Award.
- 2010 — MIT TR-35; INK Fellow; Intel IRIS Best Popular Invention.
- 2011 — NASA Award; DLF–Pramerica Spirit of Community Award; Eureka-11 IIT-Bombay.
- 2012 — Golden Book of World Record; TED-India Speaker; IDEAS-12 IIT-Kanpur.
- 2013 — Presidential Award (Pranab Mukherjee, NIF); ICAI Abu Dhabi Speaker.
- 2014 — Under-35 CEO Award.
- 2021 — STPI-Chunauti Winner.
- 2024 — ELECRAMA Winner (IEEMA); Silicon Valley Speaker.
- 2025 — CEO Club Speaker, Hyderabad.

Diplomatic & institutional engagement

BRICS Roundtables · Startup20 at G20 India · Ministry of Power · DRIIV (PSA, Govt of India) · Embassies of Italy and Mauritius · British High Commission · Indian Oil Corporation.

CHAPTER 05

Ecosystem

Six operating vehicles and a five-mark advisory roster. Independently viable. Together, a closed loop from atom to invoice.

05 · ECOSYSTEM · OPERATING GROUP

A portfolio engineered for planetary impact.

Research at Monoatom feeds materials at Grafillium, which feed industrial products at SPI. InThinks runs the intelligence layer. Starunico is the patient capital underneath. Magppie is the field deployment.

01 · 2025 · MATERIALS

Monoatom Labs

CO-FOUNDER & CEO · GRAPHENE AT SCALE

An innovative, scalable and economical method to manufacture graphene — and the applications that turn it into real-world performance gains. *monoatomlabs.com*

02 · 2025 · MATERIALS

Grafillium

CO-FOUNDER & CIO · ECO NANO ADDITIVES

Deep-tech nanomaterial additive technologies that boost efficiency and cut carbon emissions across power, logistics and heavy industry. *grafillium.com*

03 · 2024 · INDUSTRIAL SYSTEMS

SPI Industries

FOUNDER & CEO · R&D INDUSTRIAL SOLUTIONS

Innovative R&D-led industrial solutions in nanomaterial engineering — translating advanced materials science into deployable systems. *spiindustries.co*

04 · 2026 · INNOVATION STUDIO

InThinks

CO-FOUNDER · IDEATION & IP TRANSFER

An ideation and innovation studio that shapes early-stage thinking into products, then transfers or licenses the technology to partner organisations. *inthinks.com*

05 · 2026 · DEEP-TECH CAPITAL

Starunico Capital

CO-FOUNDER · MATERIALS & ENERGY

Backing founders building the materials and energy layer of the next century. *starunico.com*

06 · 2025 · DESIGN + LIVING

Magppie

CHIEF INNOVATION OFFICER · STONE WELLNESS KITCHENS

Pioneering the world's first 100% stone-built modular kitchen — transforming ordinary homes into wellness homes that protect family and planet. *magppie.com*

Advisory Roster · Five industry layers

- Vinrox · Materials
- VPRPL · Industrial Systems
- WeHear · Consumer Tech
- Tileopedia · Surface Technologies
- Magppie · Design + Living

CHAPTER 06

Voices

Institutional endorsements, on-record recognition, and editorial coverage. How sovereign bodies, foundations, global platforms, and the press have framed two decades of invention.

06 · VOICES · INSTITUTIONAL

What the institutions have on record.

GOVERNMENT OF INDIA · PRESIDENTIAL RECOGNITION

National Innovation Foundation · Six laureate cycles

"Honored by Presidents Pratibha Patil and Pranab Mukherjee across IGNITE, NIF and National Inspire platforms — among the youngest multi-cycle Presidential awardees."

MIT TECHNOLOGY REVIEW

Editorial coverage · Innovators track

"Featured in the global innovators discourse for graphene-anchored material work emerging from India's deep-tech bench."

TED · TED-INDIA

Speaker · Long-form stage

"On-stage at TED-India and allied platforms on invention, materials, and the industrial future of the carbon century."

INK TALKS · INK FELLOWS RETREAT

INK Fellow · Speaker

"Selected as an INK Fellow and featured speaker — part of a curated cohort of inventors, founders, and public thinkers."

INDIAN OIL CORPORATION

Industrial Recognition · Sujoy Choudhury, Director

"Recognized at IOCL's R&D forum for advanced-materials work at the energy interface."

News & Press · Selected Headlines

INDIA TODAY

New achiever on the block: At 20, Susant Pattnaik is a serial entrepreneur.

THE TELEGRAPH INDIA

Wonderboy innovates technological marvel — a device that can change lives of disabled people.

THE NEW INDIAN EXPRESS

Inspired to help poor techies — the teen innovator with the President's ear.

YOURSTORY

Capattery — pioneering graphene-based energy storage from India.

BUSINESS STANDARD · ANI

Game-changing innovation in battery charging technology by 6-times President awardee.

REDIFF · PTI

India hosts GraphIN 2026 to explore graphene's potential.

GOLDEN BOOK OF WORLD RECORDS

Youngest Inventor and Social Entrepreneur.

CHAPTER 07

Future Systems

Built in India. Designed for the world. Patient industrial capital, vertically integrated science, and the long arc of making Indian deep-tech globally inevitable.

07 · FUTURE SYSTEMS

Built in India. Designed for the world.

India has the talent, the demand, and the urgency to lead the carbon century. What remains is patience — the institutional patience to translate Indian invention into global industrial deployment.

These ventures, taken together, are one answer: a vertically integrated stack from atom-scale research to capital and commercialization, designed to make Indian deep-tech globally inevitable.

> *Engineer matter. Engineer capital. Engineer scale.*
> *Hold all three in one hand long enough for the work to outlive the inventor.*

The arc, in one sentence

The first decade was about inventions. The second is about systems. The third will be about handing the work to the next generation of Indian builders who never had to ask permission.

What the desk is open for

- Industrial partnerships in advanced materials.
- Capital co-architecture for deep-tech.
- Research collaboration in graphene, energy, water, or climate systems.
- Advisory and board work.
- Editorial, press, and speaking.

What is kept off the desk

Generic outreach, mass campaigns, or unsolicited fundraising decks. The inbox is small on purpose, so signal can survive.

THANK YOU · OPEN THE DIALOGUE

Let's build.

A short, specific note moves faster than a long one. One paragraph on context. One paragraph on why a conversation would change the trajectory.

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